English Medium

EDU TERIA

72nd BPSC

Prelims Online Batch

BIOLOGY

Topic wise Notes

LECTURE 19

Respiratory System- 02

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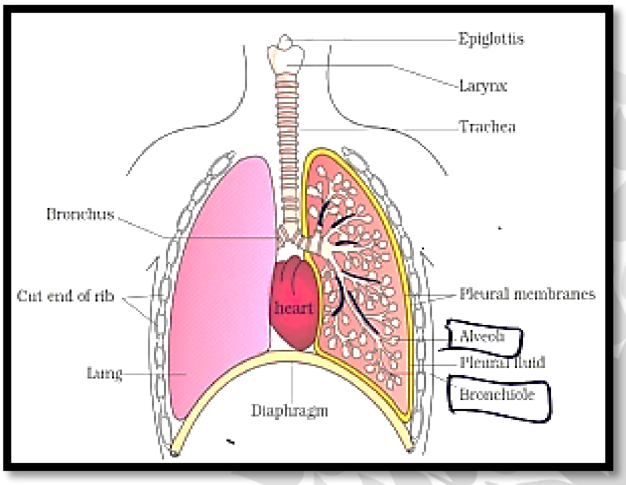
Organs of Respiratory System

1. Lungs

Lungs are the primary spongy, pinkish - gray organs of the respiratory system, located in the chest on either side of the heart, designed for breathing and gas exchange.



Lungs : Primary respiratory organ.

One pair lungs found in thoracic cavity.

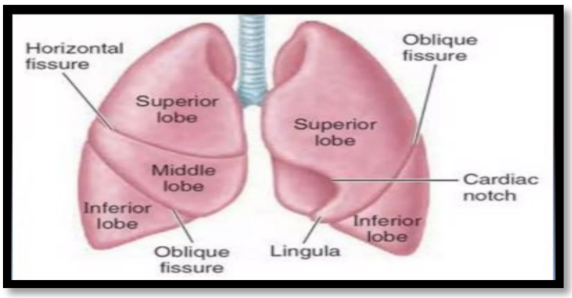
Pink spongy like structure

- Right lung–Three lobed
- Left lung– Bilobed
- Avg weight– 1-1.4 kg

Covered with double membrane layer filled with respectively fluid known as pleural membrane and pleural fluid respiratory.

- Alveoli are called as structural and functional units of lungs.
- Lungs are made up to approx. 300 million alveoli.
- Alveoli are saclike structure made up of pneumocytes cells.

Lungs divided into lobes



2. Pharynx (Throat)

Common passage for air and food

Divided into:

Nasopharynx

Oropharynx

Laryngopharynx

3. Larynx (Voice Box)

Connects pharynx to trachea

Contains vocal cords → sound production

Epiglottis prevents food entering windpipe

4. Trachea (Windpipe)

Tube supported by C-shaped cartilage rings

Carries air to lungs

Lined with cilia to trap particles

5. Bronchi & Bronchioles

Trachea divides into two bronchi

Further divide into bronchioles inside lungs

Distribute air evenly

External Respiration/Breathing

Inhalation & Exhalation combined from the process of breathing.

- Inhalation/Inspiration– Low ← High Intra pulmonary Pressure < Atmospheric pressure.
- Exhalation/Expiration– High → Low Intro pulmonary pressure > Atmospheric pressure

Mechanism of Breathing	
Inspiration	**Expiration**
Contraction in Diaphragm ↓ Ribs lift up ↓ Expansion of lungs ↓ Intrapleural pressure falls ↓ Entry of atmospheric air	Relaxation of diaphragm ↓ Ribs driven in ↓ Contraction of chest/lungs ↓ Intra pleural pressure high ↓ Exit of air

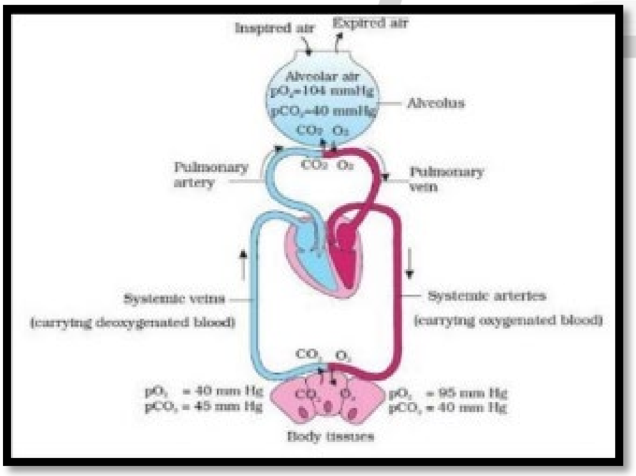
Respiratory System: Group of organs involve in exchange of gases and breakdown of glucose for energy known as respiration.

Atmospheric Air	PO_2	PCo_2
(760 mm of Hg) (in Alveoli)	Partial pressure of oxygen ↓ 104 mm of Hg	Partial pressure of Co_2 ↓ 40 mm of Hg
Air present in blood capillary	Po_2 40 mm of Hg	PCo_2 45 mm of Hg

Transportation of respiratory gases	
1. Transportation of oxygen • 97%– By Haemoglobin in th form of oxyhaemoglobin • 3%–Transported by plasma.	**2. Transportation Co_2** • 70%–In the form bicarbonate. • 23%– By haemoglobin in the form of carb amino haemoglobin. • 7%– By plasma

Cellular Respiration	
Aerobic Respiration	**Anaerobic respiration**
Breakdown of Glucose in presence of oxygen.	Incomplete breakdown of Glucose in less or absence of oxygen.
Occurs in three steps— 1. Glycolysis– Independent of oxygen 2. Link reaction– In the presence of oxygen 3. Kreb Cycle – In the presence of oxygen	Occurs only in one step i.e. Glycolysis.

Exchange of Gases



1. Glycolysis

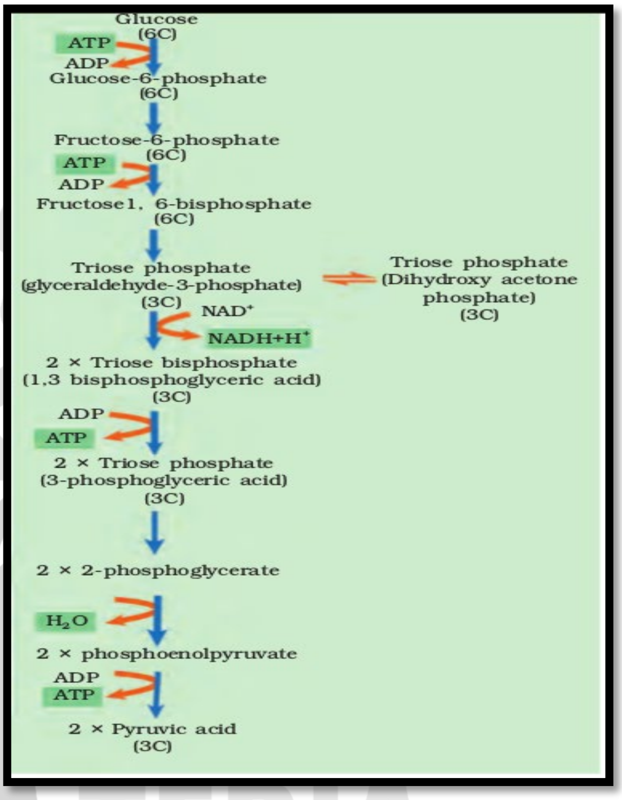
Glycolysis is the first stage of cellular respiration in which one molecule of glucose is broken down into two molecules of pyruvate. It occurs in the cytoplasm and does not require oxygen (anaerobic process).

$C_6H_{12}O_6 \rightarrow$ 2 Pyruvic Acid + 2 ATP

2. Link Reaction

Occurs in peri mitochondrial space

Pyruvic Acid $\rightarrow$ Acetyl Enzyme

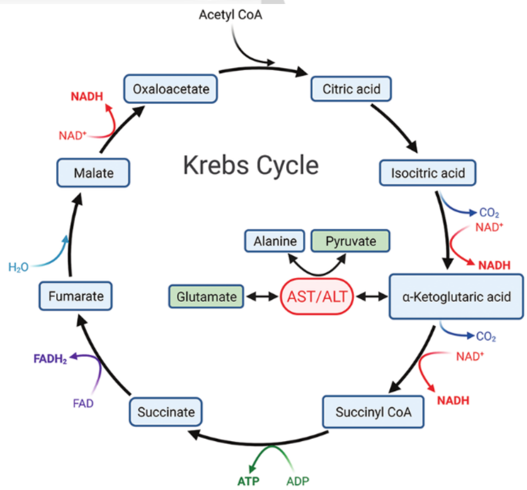


Krebs Cycle

The Krebs Cycle is the second stage of aerobic respiration, occurring in the mitochondrial matrix. It oxidizes acetyl-CoA to produce energy carriers.

Acetyl Co-enzyme $\rightarrow CO_2+H_2O$ + 24 ATP

Final Reaction – $C_6H_{12}O_6+6\ CO_2 \rightarrow 6CO_2 + 6H_2O$+38 ATP



Organism	Respiratory Organ
Unicellular animal/sponges /Tapeworm	**Body surface**
Earthworm	Cuticle/Moist skin
Insects	Trachea
Arachnida	Book lung
Fish	Gills
Pop tile	Lung
Amphibian	Skin/Gills/lung
Aves	Lung
Mammals'	Lung

Normal breathing rate — (2-16) min
Sleeping condition — 10 min
Instrument measuring among respiratory gases known as spirometer.
Normal Tidal volume = 500 ml
Residual volume = 110 -1200 ml
Volume of air always remains inside the lung.

Organism and their Respiratory Organ

	NAME OF THE ORGANISM	RESPIRATORY ORGAN		NAME OF THE ORGANISM	RESPIRATORY ORGAN
1	PROTOZOA	Respiratory organs are absent but respiration takes place by general body surface	10	MOLLUSCA	Gills (Ctenidia)
2	PORIFERA		11	ECHINODERMATA	Body surface called dermal branchiae in most, Genital bursae in Brittle star, and tube feet in all.
3	COELENTERATA		12	FISHES	Gills
4	CTENOPHORA		13	TADPOLE	Gills
5	PLATYHELMINTHES		14	FROG	Lungs /skin /buccopharyngeal cavity
6	ASCHELMINTHES		15	REPTILES	Lungs
7	ANNELIDA (EARTHWORM)	Moist cuticle	16	BIRDS	Lungs
8	Insect (eg. Cockroach, Silkworm)	Trachea	17	MAMMALS	Lungs
9.	Arachnida (eg. Spider)	Book lungs			